Phenol

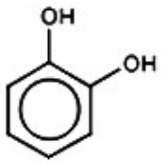
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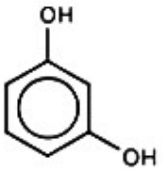
Phenol

Phenol:

Phenols are aromatic hydroxy compounds in which –OH group is connected to the benzene ring directly. Phenols are called carbolic acids. Monohydroxy benzene is phenol.
These are classified as monohydroxy, dihydroxy and trihydroxy compounds.



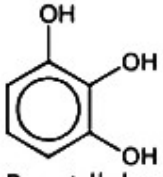
Catechol
(Benzene-1,2-diol)



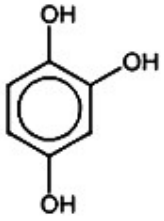
Resorcinol
(Benzene – 1, 3 – diol)



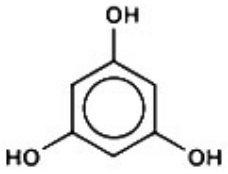
Hydroquinone
(Benzene-1,4-diol)



Pyrogallol
(1,2,3-Trihydroxybenzene)

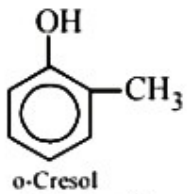


Hydroxyquinol
(1,2,4-Trihydroxy benzene)

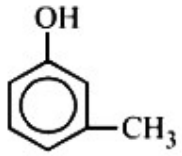


Phloroglucinol
(1,3,5-trihydroxy benzene)

Methyl phenol is called cresol. It has three position isomers.



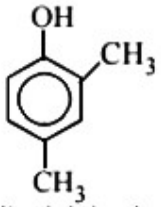
o-Cresol
(2-methyl phenol)



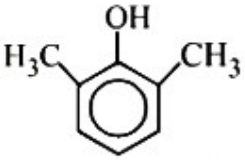
m-Cresol
(3-Methyl phenol)



m-Cresol
(4-Methyl phenol)



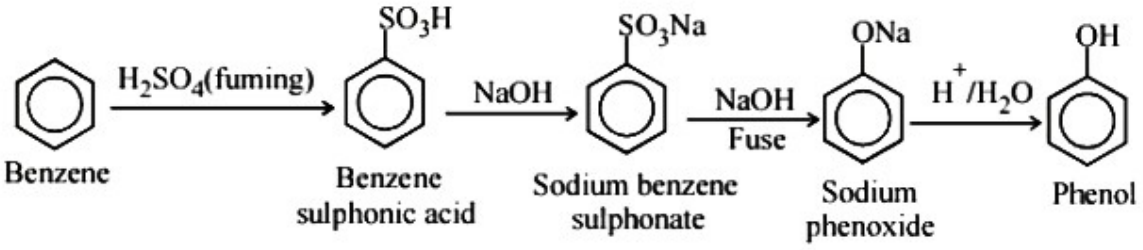
2,4-dimethyl phenol



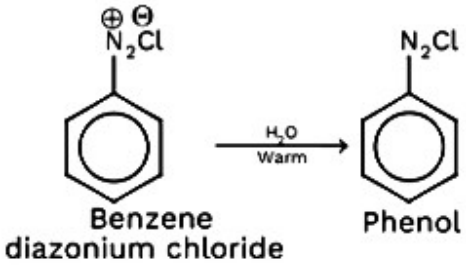
2,6-dimethyl phenol

Methods of Preparation:

(1) **From benzene sulphonic acid :** Sodium salt of benzene sulphonic acid is fused with sodium hydroxide (NaOH) when sodium phenoxide is formed. Sodium phenoxide on treatment with dilute acid yield phenol.



(2) From benzene diazonium chloride: When benzene diazonium chloride solution is warmed, phenol is formed with evolution of nitrogen. The phenol from solution is recovered by steam distillation. In this case also, benzene can be taken as the starting material. This is also a laboratory method.



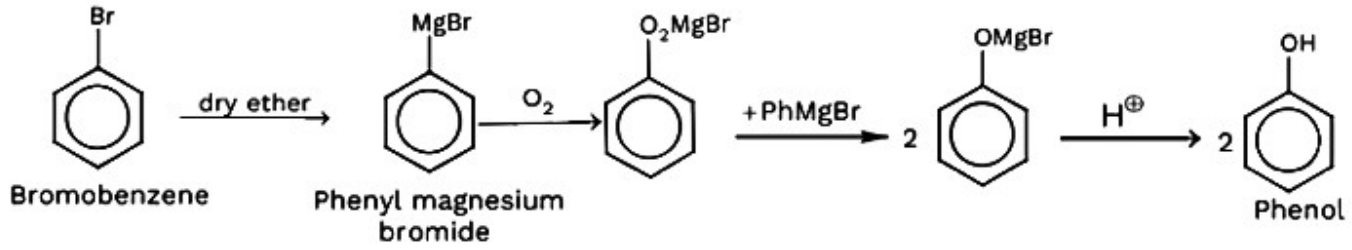
(3) From Grignard reagent:

Step - 1

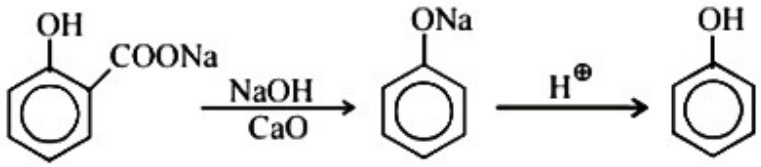
Chlorobenzene or bromobenzene is first converted into phenyl magnesium halide in presence of dry ether.

Step - 2

The Grignard reagent on reaction with oxygen and subsequent hydrolysis by a mineral acid, yields phenol.



(4) From salicylic acid: When salicylic acid or its sodium salt is distilled with soda lime, decarboxylation occurs and phenol is formed.



Industrial Preparation:

(1) Raschig's Process: Chlorobenzene is formed by the interaction of benzene vapour, hydrogen chloride and air at 250°C in presence of catalyst cupric chloride and ferric chloride. It is hydrolysed by super-heated steam at 425°C to form phenol and HCl. This is one of the latest methods for the synthesis of phenol. HCl may be again used to

convert more of benzene into chlorobenzene.

$$C_6H_6 + HCl + \frac{1}{2}O_2 \xrightarrow[250^\circ C]{CuCl_2/FeCl_3} C_6H_5Cl + H_2O$$

Benzene Chlorobenzene

$$C_6H_5Cl + H_2O \xrightarrow{425^\circ C} C_6H_5OH + HCl$$

Chlorobenzene Steam Phenol

(2) Dow process: This process involves alkaline hydrolysis of chlorobenzene. Large quantities of phenol are formed by heating chlorobenzene with a 10% solution of caustic soda or sodium carbonate at 300°C under a very high pressure (200 atm).

$$C_6H_5Cl + 2NaOH \xrightarrow[\text{High Pressure}]{300^\circ C} C_6H_5ONa + H_2O$$

Chlorobenzene

Sodium phenoxide on treatment with sulphuric acid yields phenol.

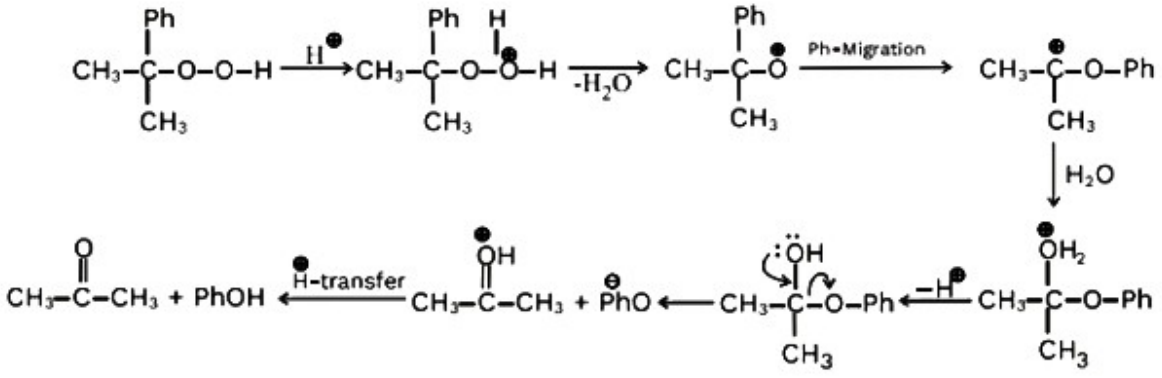
$$2C_6H_5ONa + H_2SO_4 \longrightarrow 2C_6H_5OH + Na_2SO_4$$

(3) Oxidation of benzene: This is the latest method for the manufacture of phenol. The mixture of benzene and air is passed over V_2O_5 at 315°C. Benzene is directly oxidized to phenol.

$$2C_6H_6 + O_2 \xrightarrow[315^\circ C]{V_2O_5} 2C_6H_5OH$$

(4) Oxidation of isopropyl benzene (cumene) : Cumene is oxidized to its hydroperoxide which is decomposed by dilute sulphuric acid into phenol and acetone.

The mechanism of the conversion of the cumene hydroperoxide into phenol is not certain, but it most probably involves a 1,2 shift of Ph from C to O.



Physical Properties:

(a) Phenol is a colourless crystalline, deliquescent solid. It turns pink colour on exposure to air and light.

(b) Its melting point is 43°C and boiling point 182°C.

Q1 **o-Nitrophenol has a lower b.p and lower water solubility than the p-isomer. Explain.**

Sol: In the o-compound, the hydrogen bonding is intramolecular, and so the molecule behaves as a single unit. Hydrogen bonding in the p- compound is intermolecular. Thus, we are dealing with related molecules, which behave as a large molecule with high molecular weight, thereby giving rise to a high boiling point (much higher than that of the monomer)
Solubility depends on a number of factors, one being the ability of the solute to form hydrogen bonds with the solvent, which is water in this case. Because the intramolecular hydrogen bonding in the o-compound is (apparently) unaffected by water, little hydrogen bonding occurs between the two, and so the o-compound shows tends to undergo disassociation into the monomeric molecules and these can now form hydrogen bonds with the water, hereby having higher degree of solubility.

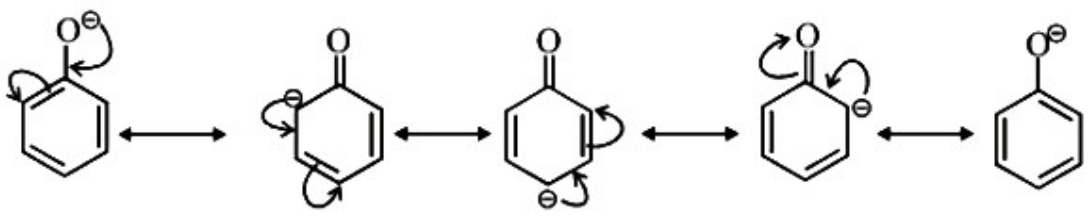
Chemical Properties:

1. Acidic Nature

$C_6H_5 + H_2O \rightleftharpoons C_6H_5O^- + H_3O^+$

Phenoxide ion

The phenoxide ion is steady due to resonance.



The negative charge is spread throughout the benzene ring. This charge delocalisation is a stabilizing factor in the phenoxide ion and grow acidic nature of phenol. No resonance is imaginable in alkoxide ions (RO–) derived from alcohols. The negative charge is contained on oxygen atom. Thus, alcohols are less acidic than phenol.

Effect of substituents on the acidity of phenols:

Presence of electron attracting group (e.g., $-HO_2$,$-X$,$-NR^+_3$,$-CN$,$-CHO$,$-COOH$) on the benzene ring growths the acidity of phenol as it enables the ring to draw more electrons from the phenoxy oxygen and therefore releasing easily the proton. The particular effect is added when the substituent present on o- or p-position than in m-position to the phenolic -OH group.

The relative acidic strengths of some phenols are as follows:
p-Nitrophenol > o-Nitrophenol > m-Nitrophenol > Phenol.

Presence of electron releasing group, (e.g., $-CH_3$,$-C_2H_5$) on the benzene ring decreases the acidity of phenol as it strengthens the negative charge on phenoxy oxygen and thus proton release becomes difficult. Thus, cresols are less acidic than phenol.

Examples of acidic nature of phenols:

Chlorophenols	-	o- > m- > p- > phenol
Cresols	-	m- > p- > o-
Dihydric phenols	-	m- > o- > p-
Methoxy phenols	-	m- > o- > p-
Amino phenols	-	o- > m- > p-

Points to remember

Greater the value of K_a or lower the value of pK_a, stronger will be the acidity.

Observation of phenol test :

The acidic nature of phenol is observed in the following:

(i) Phenol changes blue litmus to red.

(ii) Highly electropositive metals react with phenol.

$$2C_6H_5OH + 2Na \longrightarrow 2C_6H_5ONa + H_2$$

(iii) Phenol reacts with strong alkalies to form phenoxides.

$$C_6H_5OH + NaOH \longrightarrow C_6H_5ONa + H_2O$$

Though, phenol does not decompose sodium carbonate or sodium bicarbonate, i.e., CO_2 is not evolved because phenol is weaker than carbonic acid.

Q2 **Arrange on the basis of decreasing acidic strength of given substituted phenols:**
(a) Phenol, o-nitrophenol, m-nitrophenol, p-nitrophenol
(b) Phenol, p-chlorophenol, 2,4,6- trichlorophenol, 2,4 dichlorophenol
(c) Phenol, benzyl alcohol, benzenesulfonic acid, benzoic acid

Sol: (a) Intramolecular H-bonding makes the o-isomer weaker than the p-isomer. The decreasing order of relative acids strengths p-nitrophenol > o-nitrophenol > m-nitrophenol > phenol.
(b) The decreasing order of relative acid strengths 2,4,6-trichlorophenol > 2, 4- dichlorophenol > p-chlorophenol > phenol
(c) The decreasing order of relative acid strengths benzene sulphonic acid > benzoic acid > phenol > benzyl alcohol

2. Reactions of OH group:

(i) **Reaction with $FeCl_3$:** Phenol gives violet colour with neutral $FeCl_3$ due to the formation of a coloured iron complex, which is a characteristic to the existence of keto-enol tautomerism in phenols (predominantly enolic form).

(ii) **Ether formation:** Phenol reacts with alkyl halides in alkali medium to produce phenyl ethers. The phenoxide ion is a nucleophile, will trade halogen of alkyl halide.

$$C_6H_5OH + NaOH \longrightarrow \underset{\text{Sod. phenoxide}}{C_6H_5ONa} + H_2O$$

$$C_6H_5ONa + CH_3I \longrightarrow \underset{\substack{\text{Methyl phenyl}\\ \text{ether (Anisole)}}}{C_6H_5OCH_3} + NaI$$

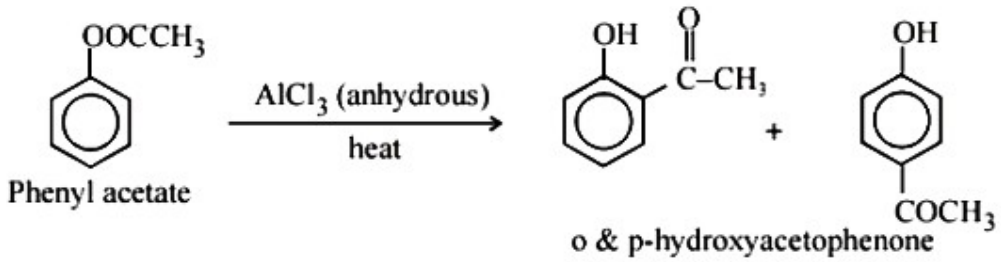
$$C_6H_5OK + C_2H_5I \longrightarrow \underset{\text{Ethoxy benzene}}{C_6H_5 - O - C_2H_5} + KI$$

$$\underset{\text{Isopropyl iodine}}{C_6H_5ONa + (CH_3)_2CHI} \longrightarrow \underset{\text{Isopropyl phenyl ether}}{C_6H_5 - O - HC(CH_3)_2} + NaI$$

(iii) **Ester formation :** Phenol reacts with acid chlorides (or acid anhydrides) in alkali solution to form phenyl esters (Acylation).

$$C_6H_5OH + (CH_3CO)_2O \xrightarrow{CH_3COONa} CH_3COOC_6H_5 + CH_3COOH$$

The phenyl esters on action with anhydrous $AlCl_3$ undergoes Fries rearrangement to give o- and p- hydroxy ketones.

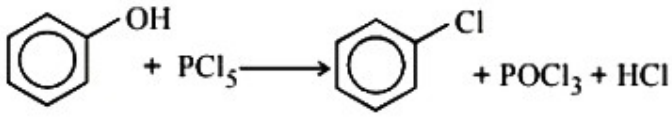


Phenyl acetate

o & p-hydroxyacetophenone

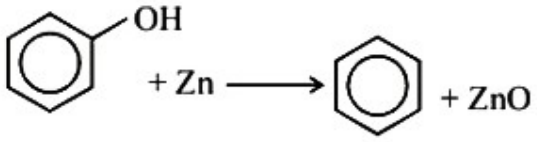
(iv) Reaction with PCl_5

Phenol reacts with PCl_5 to form chlorobenzene. The yield of chlorobenzene is very poor and mainly triphenyl phosphate is formed.

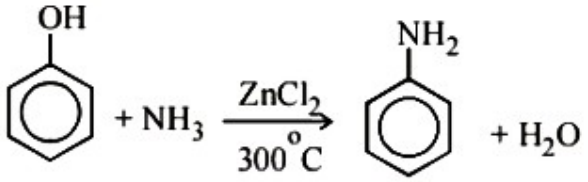


$$3C_6H_5OH + POCl_3 \longrightarrow (C_6H_5)_3PO_4 + 3HCl$$

(v) Reaction with zinc dust : When phenol is distilled with zinc dust, benzene is obtained.



(vi) Reaction with ammonia : Phenol reacts with ammonia in presence of anhydrous zinc chloride at 300°C to form aniline.

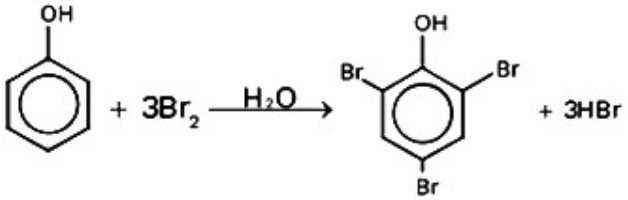


Aniline

(3) Reactions of benzene ring :

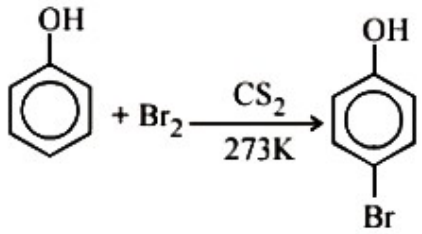
The –OH group is ortho and para directional and activates the benzene ring.

(i) Phenol forms a white precipitate with excess of bromine water yielding 2,4,6-tribromophenol.



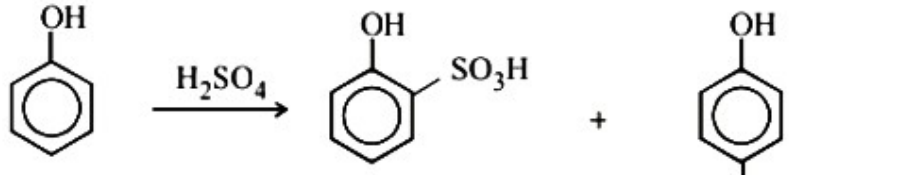
2,4,6-Tribromophenol

(ii) **Halogenation :** Phenol reacts with bromine in carbon disulphide (or $CHCl_3$) at low temperature to form para bromophenol.



p-Bromophenol

(iii) **Sulphonation :** Phenol reacts with conc. Sulphuric acid readily to form mixture of ortho and para hydroxy benzene sulphonic acids.

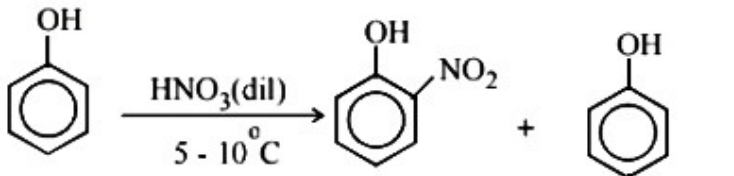


o-Hydroxybenzene sulphonic acid

p-Hydroxybenzene sulphonic acid

At low temperature (25°C), the ortho-isomer is the major product, whereas at 100°C it gives mainly the para-isomer.

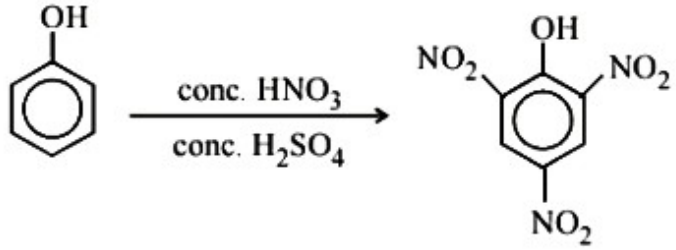
(iv) **Nitration:** Phenol reacts with dilute nitric acid at 5 – 10°C to form ortho and para nitro phenols . Yield is less due to oxidation of phenolic group. The hydroxy group is activating group, hence nitration is possible with dilute HNO_3.



o-Nitrophenol

p-Nitrophenol

When phenol is treated with conc. nitric acid in presence of concentrated H_2SO_4 compound (picric acid) 2,4,6-trinitrophenol is designed.

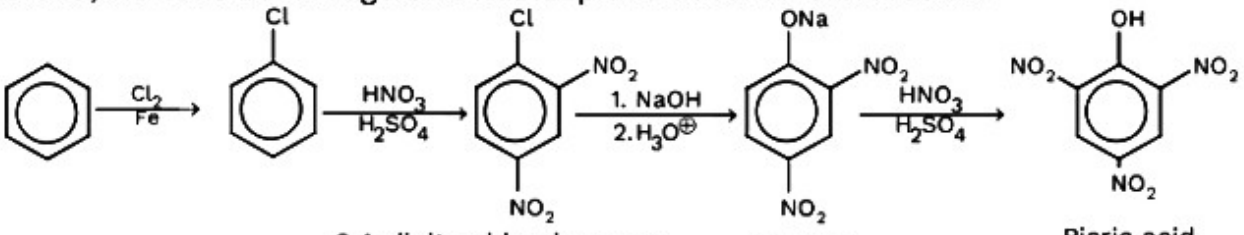


2,4,6-Trinitrophenol
(picricacid)

To obtain better yield of picric acid, very first sulphonation of phenol is made and then nitrated. Presence of $-SO_3H$ group prevents oxidation of phenol.

Q3 **Prepare (a) Picric acid 2,4,6- trinitrophenol, and (b) 2,4-dichlorophenol from benzene.**

Sol: Even though the NO_2 groups are in the o & p-positions where –OH directs electrophilic substitution, phenol cannot be nitrated beccuase the ring is susceptible to oxidative ring cleavage by nitric acid. Instead, we take advantage of a nucleophilic addition elimination.

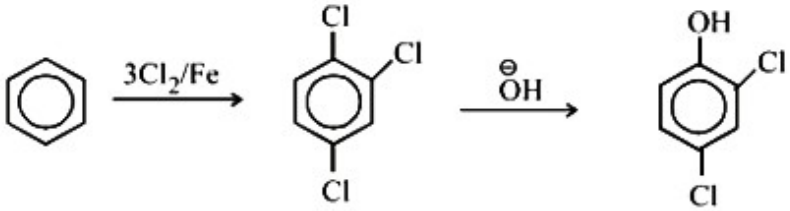


2,4-dinitrochlorobenzene

Sodium
2,4-dinitrophenolate

Picric acid

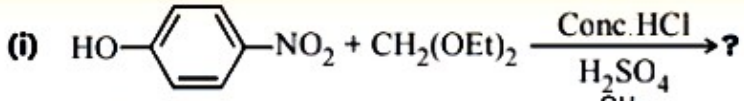
Chlorobenzene cannot be trinitrated because the Cl and two NO_2,s deactive the ring toward further electrophilic substitution.2,4-dinitrophenol can be nitrated because the two deactivating NO_2,s prevent ring oxidation. (b) Phenol cannot be chlorinated because the ring is susceptible to oxidation by Cl_2. Again nucleophilic addition – elimination is used.

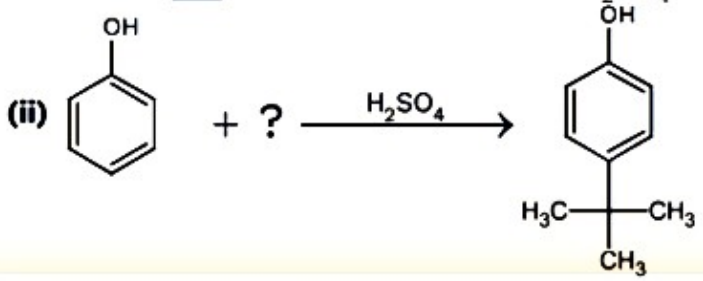


1,2,4-Trichlorobenzene

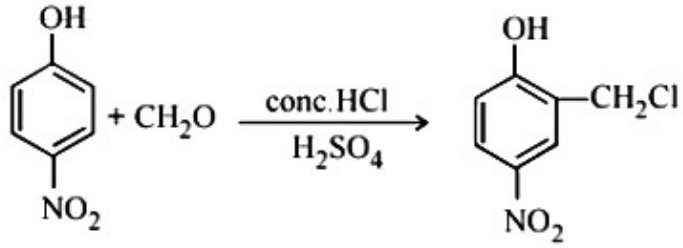
2,4-Dichlorophenol

The displaced Cl is ortho to one other Cl and para to the other and is activated by both. Each of the other Cl's is meta to at least one deactivating Cl.

Q4 (i) 

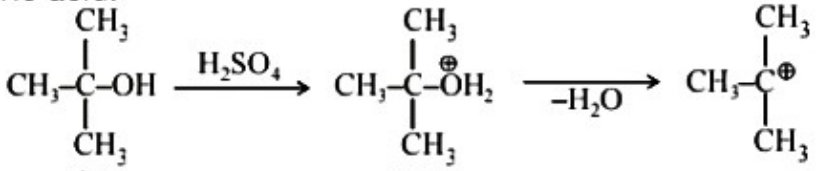
(ii) 

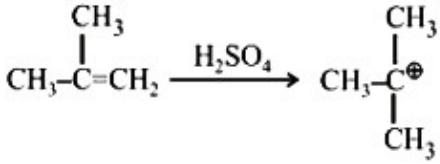
Sol: (i) $CH_2(OEt)_2 \xrightarrow[H_2O]{HCl} CH_2O + 2EtOH$

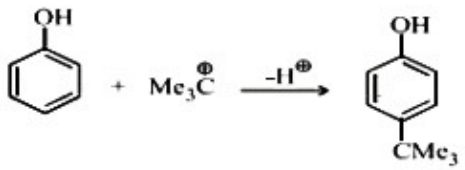


This is an example of chloromethylation; the presence of the nitro-group permits the isolation of the monochloromethyl derivative.

(ii) This is an alkylation reaction (Friedel-Crafts reaction), and since sulphuric acid is the catalyst, the reactive species is a carbonium ion. Also, because the ion is t-butyl, this can be conveniently generated in two ways with sulphuric acid.

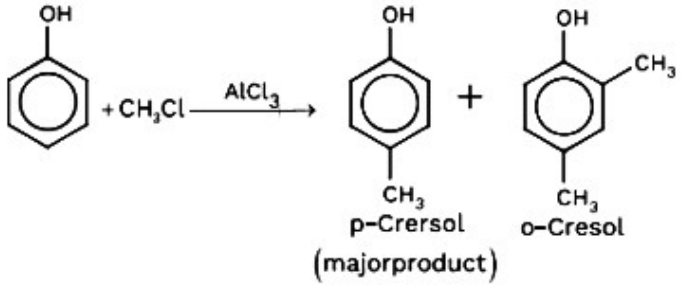
(a) 

(b) 

Then : 

This is an example of the dienone-phenol rearrangement. The methyl group migrates to the o-position, in this case to the one para to the nuclear methyl group because of the steric effect.

(v) Friedel-Craft's Reaction: Phenol when treated with methyl chloride in presence of anhydrous $AlCl_3$, p-cresol is the main product. A very small amount of o- cresol is also formed.



To obtain successful Friedel-Craft's reaction with phenol it is necessary to use a large amount of $AlCl_3$.

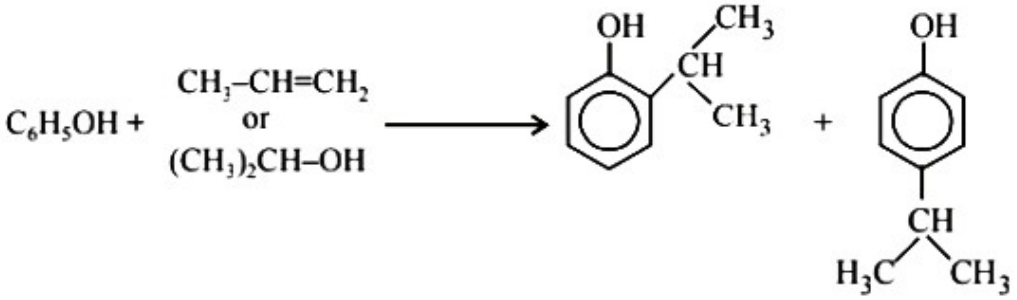
The **Ring alkylation** takes place as follows:

Example :

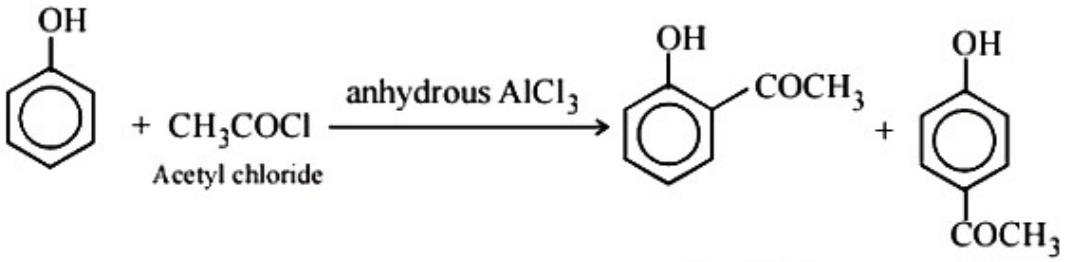
Points to remember

RX and $AlCl_3$ give poor yields because $AlCl_3$ co-ordinates with O. So Ring alkylation takes place as follows :

$C_6H_5OH + AlCl_3 \longrightarrow C_6H_5OAlCl_2 + HCl$



Example :

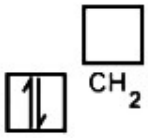
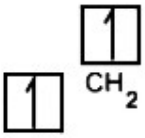
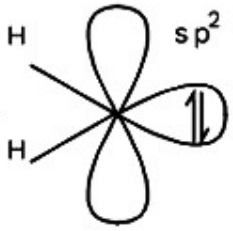
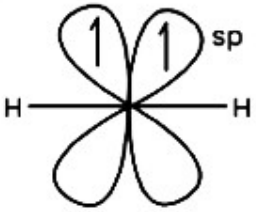


o & p-Hydroxy acetophenone

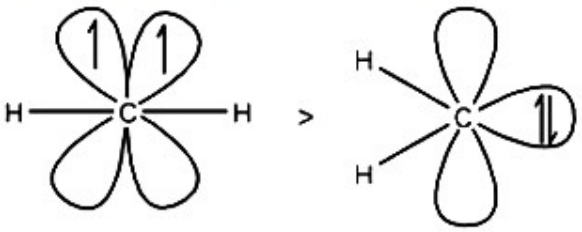
Carbenes :

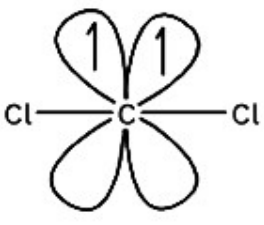
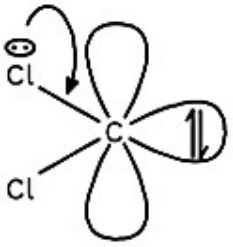
Carbenes, bivalent carbon species, are well-known sensitive intermediates that enable new carbon–carbon bond formation. A characteristic carbene reaction is the addition to a π bond of olefins. Carbenes exhibit diverse reactivity. Singlet carbenes possess equally electrophilic and nucleophilic character, while triplet carbenes show diradical reactivity.

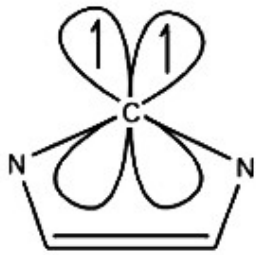
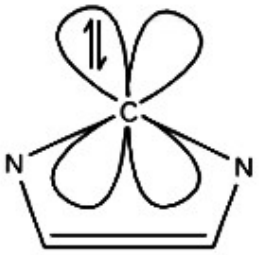
Types of Carbene (Singlet and Triplet Carbenes)
A carbene is a neutral bivalent carbon species comprising two electrons that are not shared by other atoms. When these two electrons have conflicting spins, the carbene is selected a singlet carbene; when they take similar spins, the carbene is a triplet. In the ground state, a singlet carbene takes a couple of electrons in a single orbital, while the triplet has two unpaired electrons, each inhabiting a separate orbital.

Singlet	**Triplet**
1. 	
2. 	
3. Electrophile	Bi Radical
4. Diamagnetic	Paramagnetic
5. No Reaction with O_2	Reactive with O_2

Stability in carbenes :
In general stability of triplet is more compare to singlet still in some cases due to extra factors singlet may be more stable.

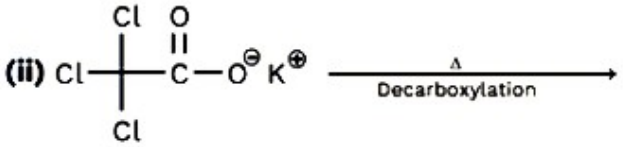
1. 

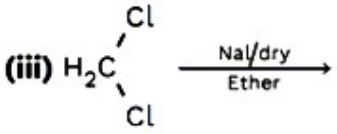
2.   Backbond

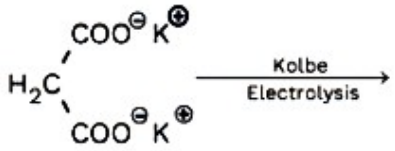
3.  <  Reso stabilised

Q5 **Suggest below reactions mechanism for preparation of carbene :**

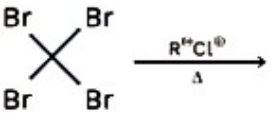
(i) $CHCl_3 \xrightarrow{KOH}$

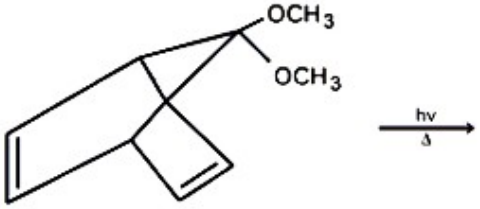
(ii) 

(iii) 

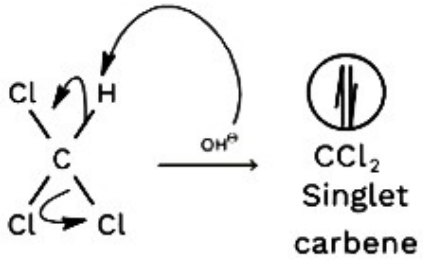
(iv) 

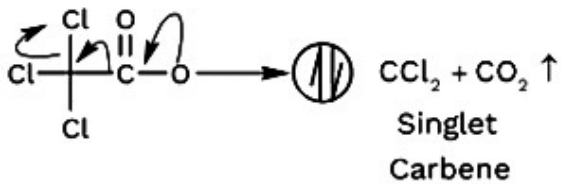
(v) $\overset{\ominus}{C}H_2 - \overset{\oplus}{N} \equiv N \xrightarrow{\Delta}$

(vi) 

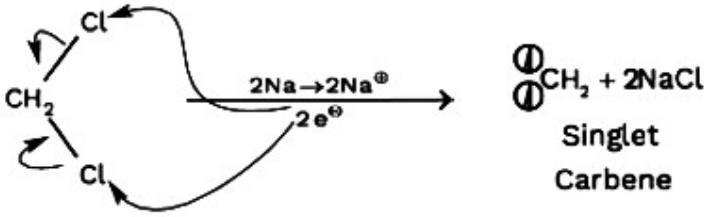
(vii) 

Sol:

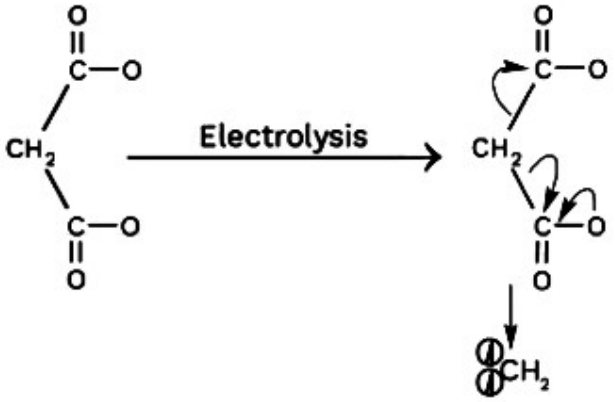
(i) 

(ii) 

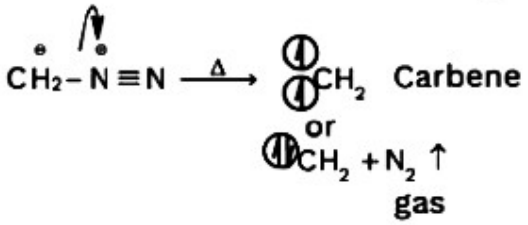
(iii)



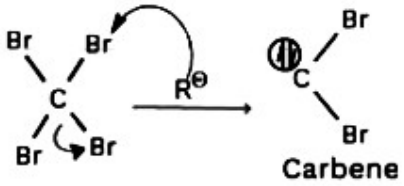
(iv)



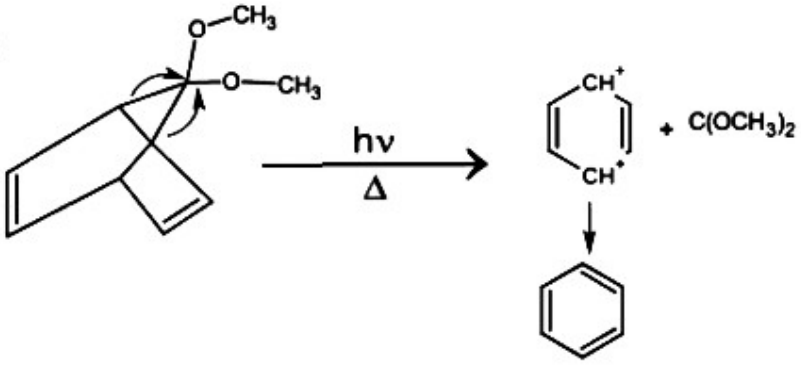
(v)



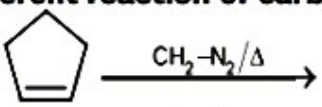
(vi)



(vii)



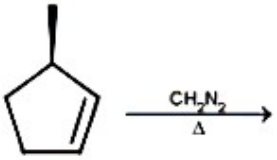
Q6 **Different reaction of carbene addition mechanism**

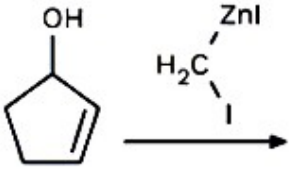
1. 

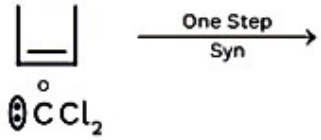
2. $CH_2 = C = CH_2 \xrightarrow[CH_2N_2]{\text{2 mole}}$

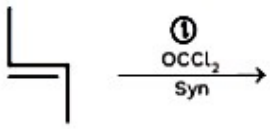
3. $CH \equiv CH \xrightarrow[CH_2N_2]{\text{2 mole}}$

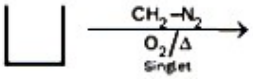
4. $Ph - C \equiv C - Ph \xrightarrow[KOH]{CHCl_3}$

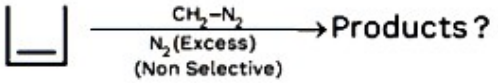
5. 

6. 

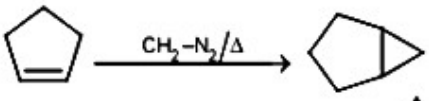
7. 

8. 

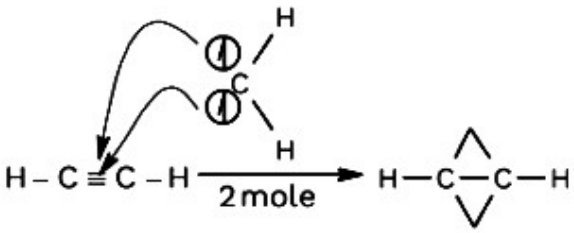
9. 

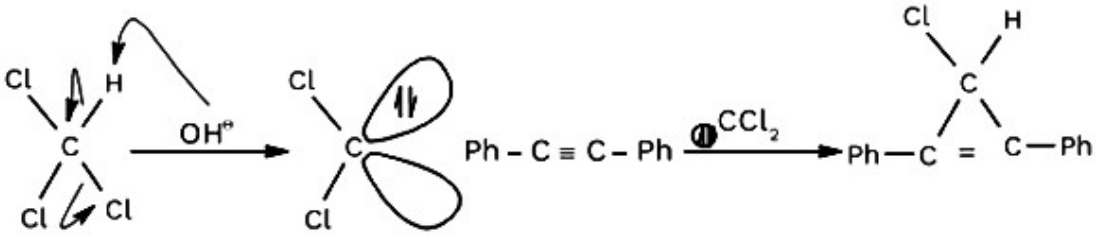
10. 

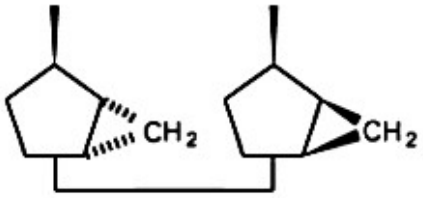
Sol:

1. 

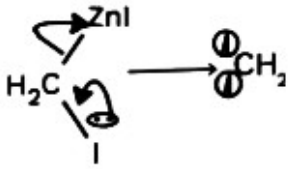
2. $CH_2 = C = CH_2 \xrightarrow[CH_2N_2]{\text{2 mole}}$  CH_2

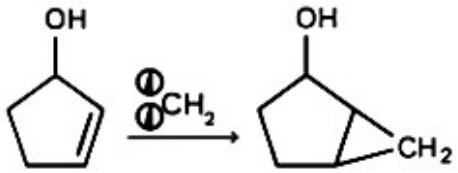
3. 

4. 

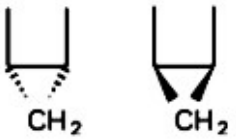
5. 

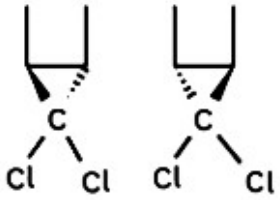
Diastereo – isomer

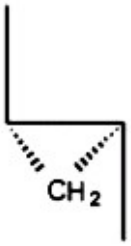
6. 



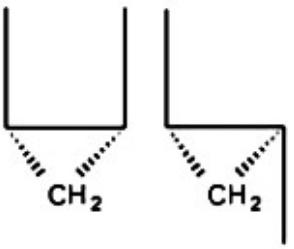
Simmon Smith Reaction

7. 

8. 

9. 

Conformation possible

10. 

Q7 **Stability order of following singlet halocarbene is**

(A) $CF_2 > CCl_2 > CBr_2 > Cl_2$

(B) $Cl_2 > CBr_2 < CCl_2 > CF_2$

(C) $CCl_2 > CF_2 < CBr_2 > Cl_2$

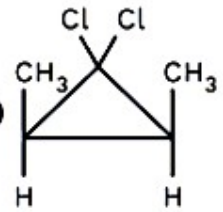
(D) $CF_2 > Cl_2 > CCl_2 > CBr_2$

Sol: **A**

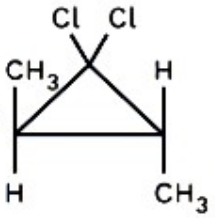
Stability is proportional to extent of back bonding by -X

Q8 **Trans-Butene-2** $\xrightarrow[\text{Solvent}]{CHCl_3/KOH}$ **Product**

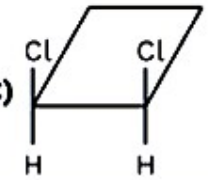
(A)



(B)



(C)



(D) Both A and B

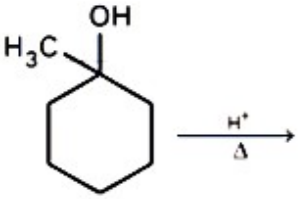
Sol: **B**

Q9 **Match the column:**

Column I		Column II	
(A)	Electrophile	(P)	$CH_3 - CH = CH - CH_3$
(B)	Nucleophile	(Q)	$:CHCl$
(C)	Lewis acid	(R)	$CH_3 - \dot{C}H_2$
(D)	Lewis base	(S)	Guanidine

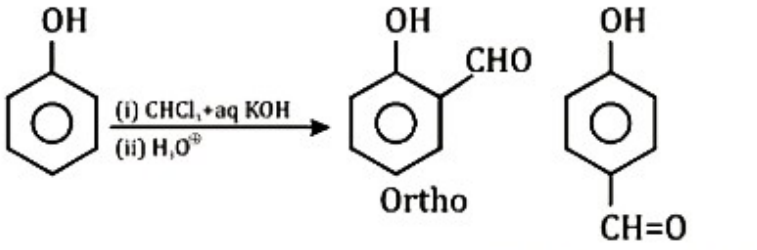
Sol: **A – Q, R;** **B – P, S;** **C – Q;** **D – P, S**

Q10

Column I (Reaction)	Column II (Type of intermediate formed)
(A) $Ph-CHCl_2 \xrightarrow[\Delta]{HO^{\ominus}} (A)$	(P) Carbocation
(B) $R-Br \xrightarrow[\text{dry ether}]{Na}$	(Q) Carbanion
(C) $H_3C-\overset{\overset{O}{\|}}{C}-CH_3 \xrightarrow[(ii)\,H_2SO_4\,\Delta]{(i)\,Mg/H_2O}$	(R) Free – radical
(D) 	(S) Carbene

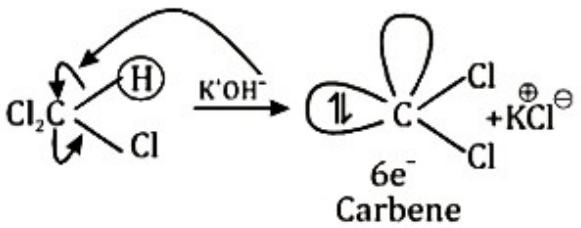
Sol: **A – Q, S;** **B – Q, R;** **C – P, R** **D – P**

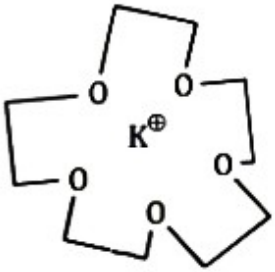
(vi) Reimer-Tiemann reaction : Phenol on refluxing with chloroform and sodium hydroxide (aq.) followed by acid hydrolysis yields salicyladehyde (o-hydroxy benzaldehyde) and a very small quantity of p-hydroxy benzaldehyde. However, when carbon tetrachloride is castoff, salicylic acid (predominating product) is formed.



Para Salicyaldehyde

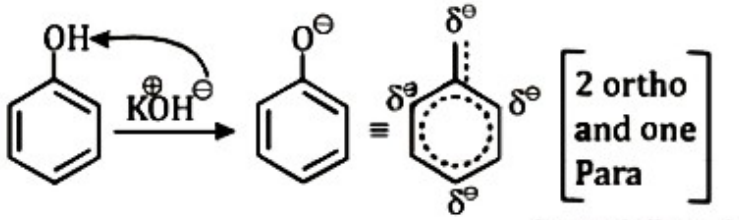
Mechanism:





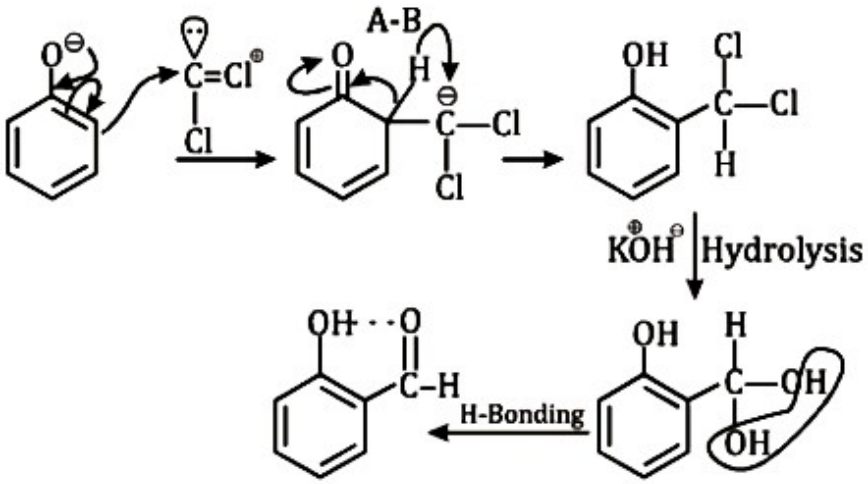
* Guest–Host Pair

Step I :



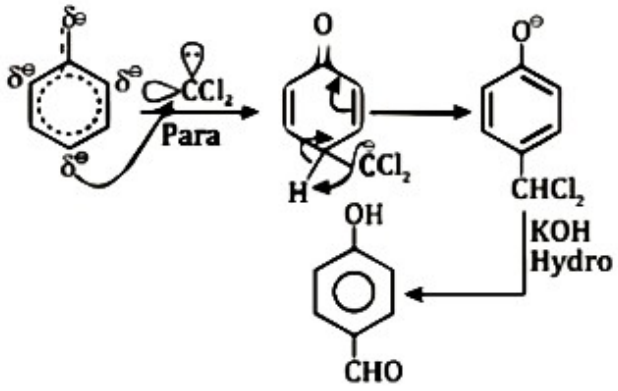
(More Collision)

Step II :



B.P.⁻(Intramolecular)

Para product :



Characteristics of reaction:

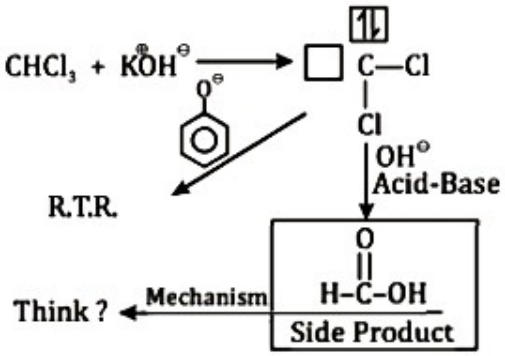
(a) In R.T.R. carbene and carbanion formed as intermediate.

(b) In R.T.R. ortho is major product because its having two reacting sites. as well as a product with intramolecular H-bonding

(c) In H_2O, oxygen is less nucleophilic as it engaged in H-bonding.

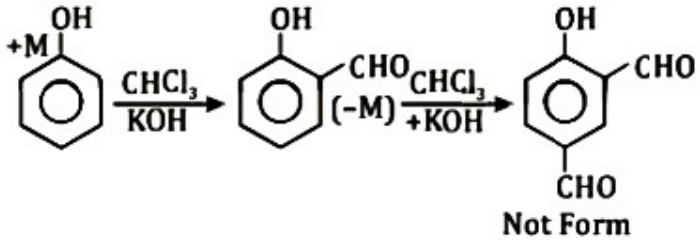
(d) Separation of mixture is easy because boiling point of ortho is less due to inter molecular H-bonding.

(e) In R.T.R. product yield is less due to side reaction.

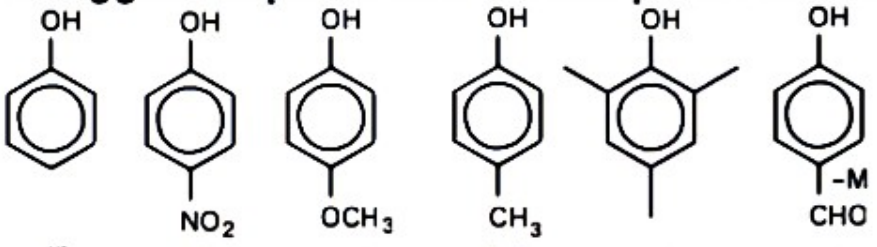


Side Product → Formic Acid (50%)

(f) Reimer tiemann reaction takes place only one time because poor NU^- can't with poor electrophilic.



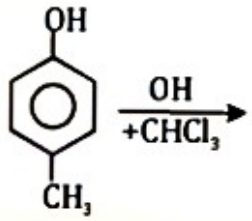
Q11 **Among given compounds which will show positive R.T.R. reaction?**



Sol: **(i), (ii), (iii), (iv), (v)**

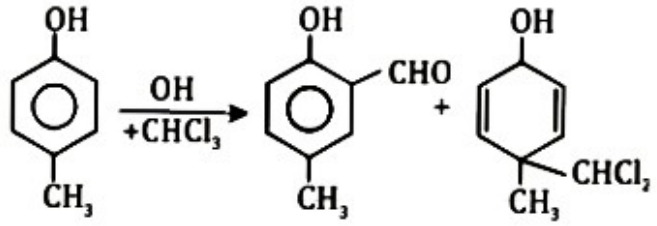
Due to presence of -m groups and hence poor nucleophilicity of ring (ii) and (vi) do not react

Q12

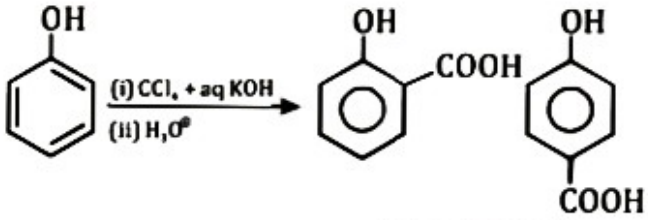


[IIT-2012]

Sol:

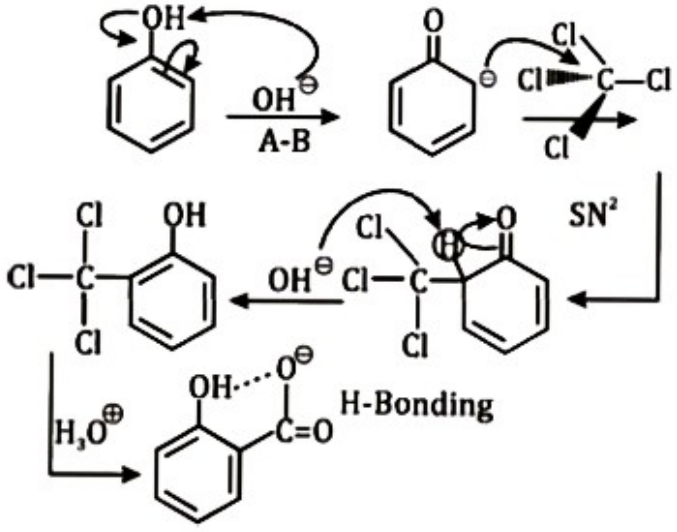


(g) In R.T.R. if chloroform is replaced by CCl_4 then it is known as Reimer tiemann carboxylation.



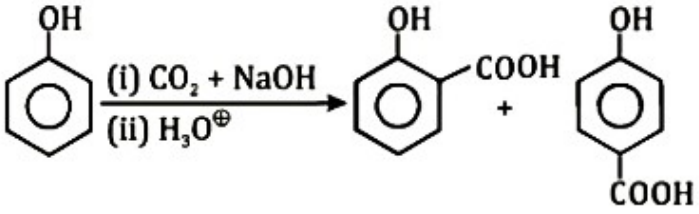
Salicylic Acid

Mechanism:

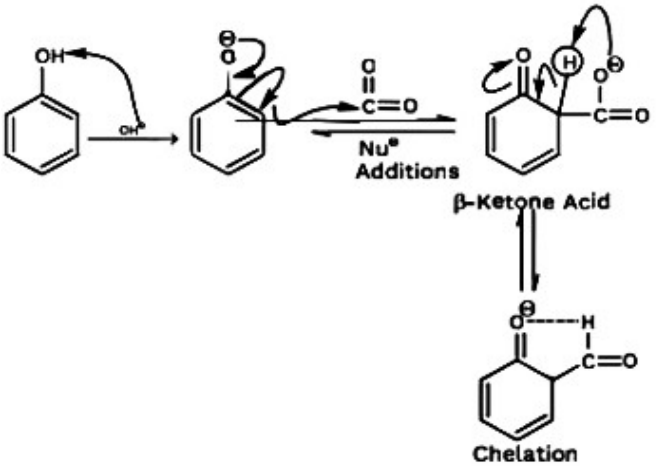


(vii) Kolbe - Schmidt Reaction :

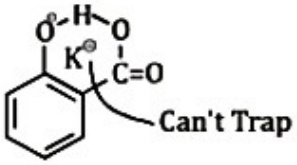
This involves the interaction of sodium phenoxide with carbon dioxide at 130-140°C under pressure of 6 atm followed by acid hydrolysis. Salicylic acid (o-Hydroxy benzoic acid) is formed.



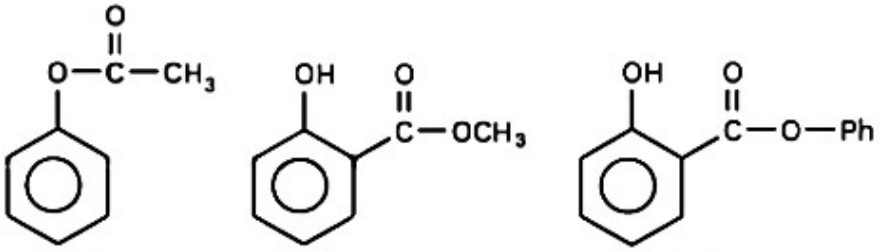
Mechanism



NaOH : Ortho Major
KOH : Para Major



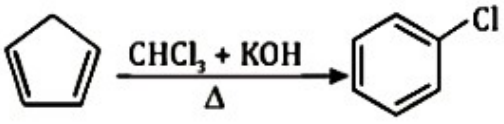
In Kolbe smith, ortho is major in presence of NaOH because of chelation but in presence of KOH, para is major because of large size of K+ Chelation not possible at ortho isomer and due to e– repulsion reaction in backward direction.



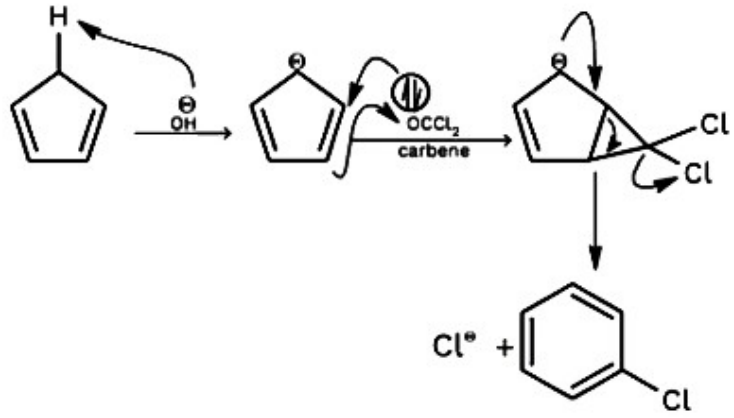
Aspirin Iodex Salol

*** If phenol is replaced by cyclopentadiene then it is known as abnormal R.T.R.**

Example:



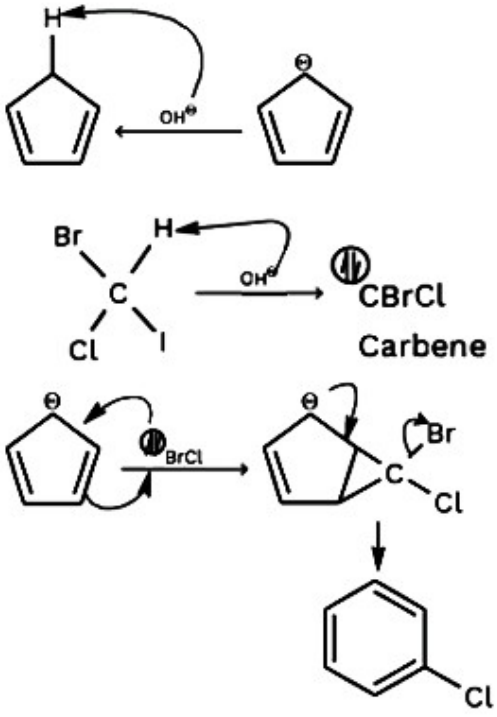
Mechanism??



Q13



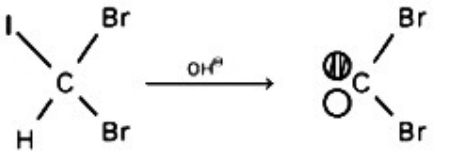
Sol:

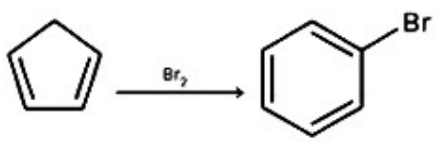


Q14



Sol:

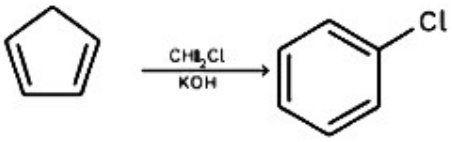




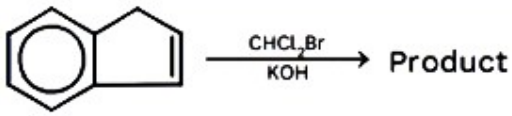
Q15



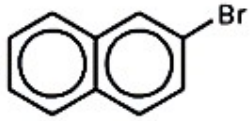
Sol:

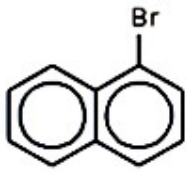


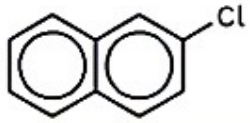
Q16

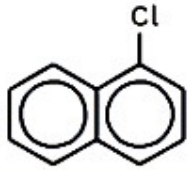


Product is :

(A) 

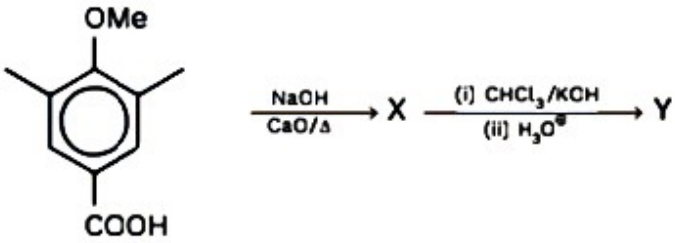
(B) 

(C) 

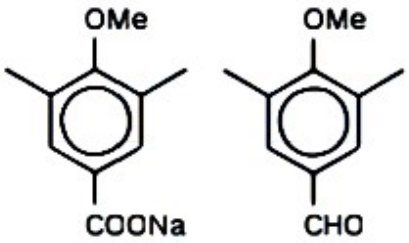
(D) 

Sol: **(c)**

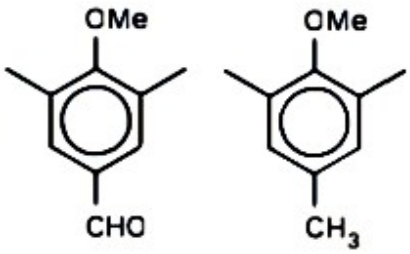
Q17 **Identify X and Y :**



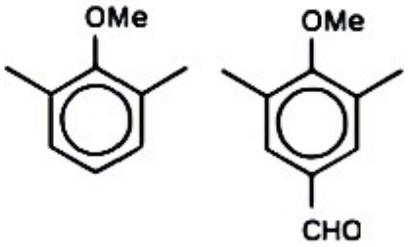
(A)



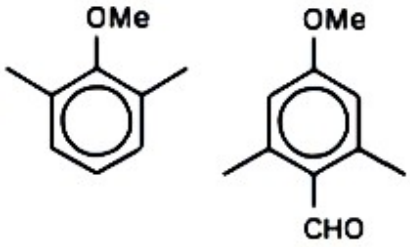
(B)



(C)

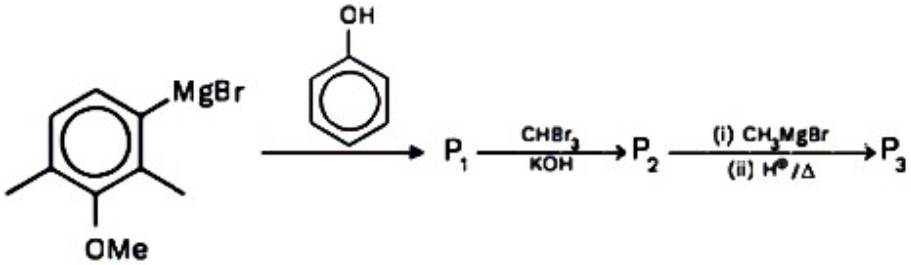


(D)



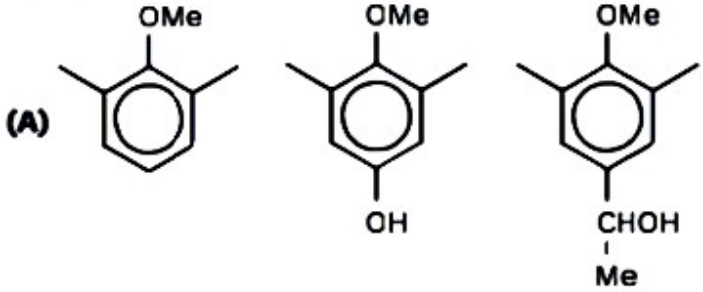
Sol: **(c)**

Q18

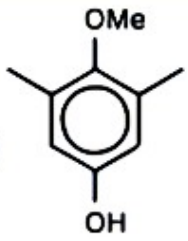


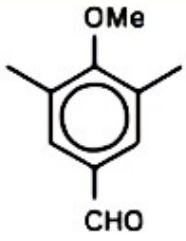
P_1, P_2 and P_3 are

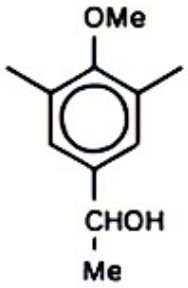
(A)



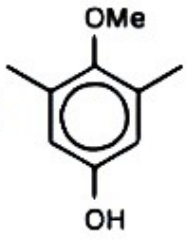
(B)

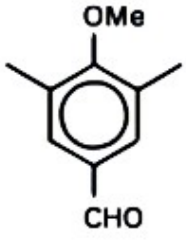


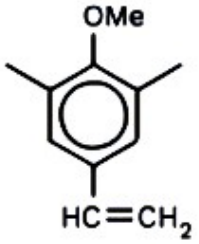




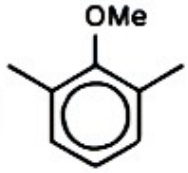
(C)

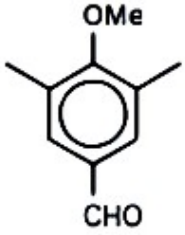


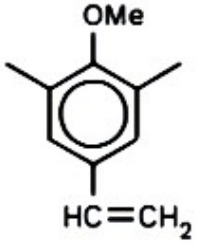




(D)







Sol: **(D)**

Q19 **Select the correct statements :**

(A) Protonation increases electrophilic nature of carbonyl group

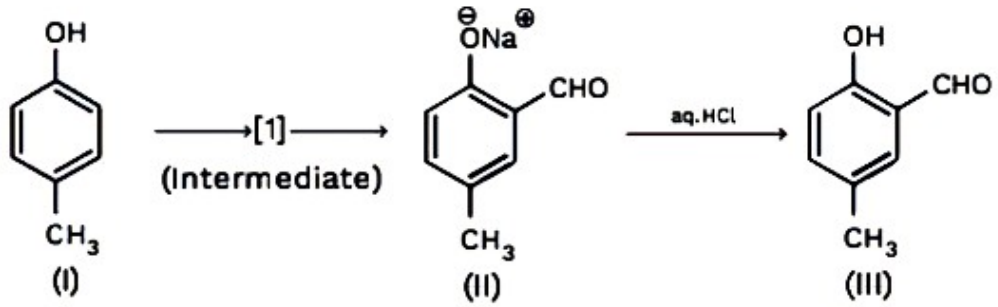
(B) $CF_3SO_3^-$ is better leaving group than $CH_3SO_3^-$

(C) Benzyl carbocation is stabilized by resonance

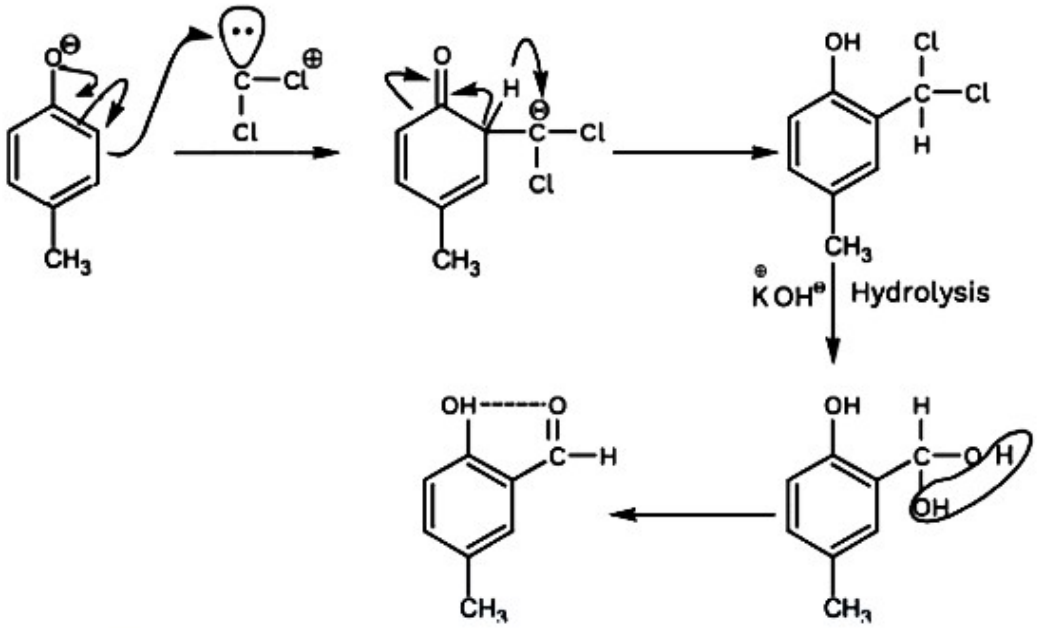
(D)  **is unstable**

Sol: **(ABCD) All statements are correct**

Q20 **Riemer Tiemann reaction introduces an aldehyde group, on to the aromatic ring of phenol, ortho to the -OH group. This reaction involves electrophilic aromatic substitution. This is a general method for the synthesis of substituted salicylaldeydes as depicted below.**



Sol:



Q21 **Which one of the following regents is used in the above reaction ?** **[JEE 2007]**

(A) aq. NaOH + CH_3Cl **(B) aq. NaOH + CH_2Cl_2**
(C) aq. NaOH + $CHCl_3$ **(D) aq. NaOH + CCl_4**

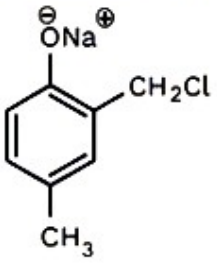
Sol: **(C)** aq. NaOH + $CHCl_3 \longrightarrow :CCl_2 + NaCl + H_2O$

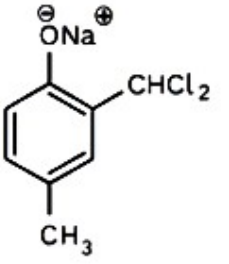
Q22 **The electrophile in this reaction is** **[JEE 2007]**

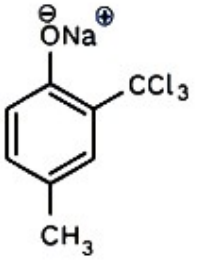
(A) :CHCl **(B) $^{\bullet}CHCl_2$**
(C) :CCl_2 **(D) $^{\times}CCl_3$**

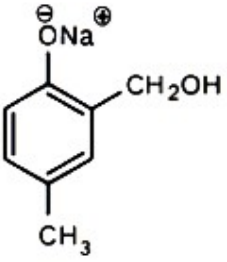
Sol: **(C)** aq. $NaOH + CHCl_3 \longrightarrow :CCl_2 + NaCl + H_2O$

Q23 **The structure of the intermediate I is :**

(A)


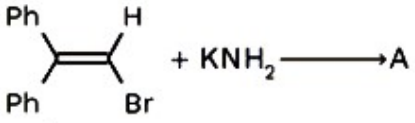
(B)


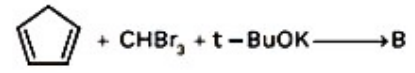
(C)


(D)


Sol: **(B)**

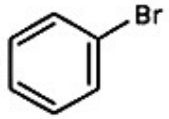
Q24 **Complete the following giving the structures of the principal organic products :** **[Jee 1997]**

(A)


(B)


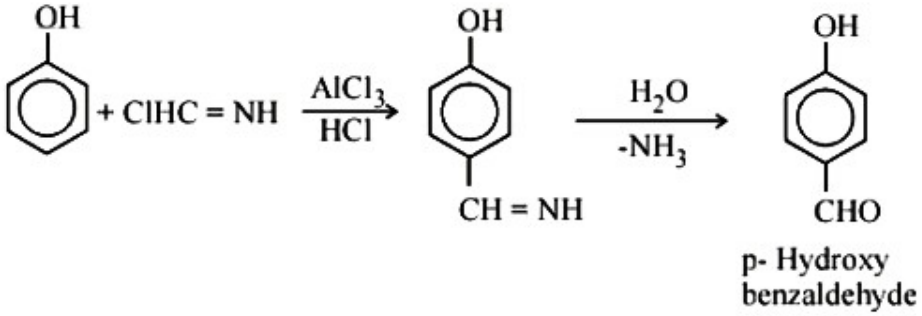
Sol:

(A) $PhC \equiv CPh$,

(B)


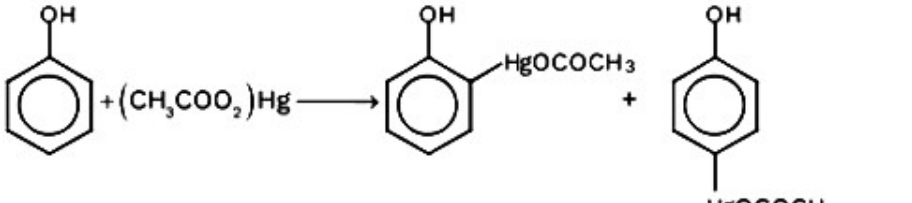
(viii) Gattermann's reaction : Phenol when treated with liquid hydrogen cyanide and hydrochloric acid gas in presence of anhydrous aluminium chloride yields mainly p-hydroxy benzaldehyde (Formylation).

$$HCl + HC\equiv N \xrightarrow[HCl]{AlCl_3} ClHC = NH$$



p- Hydroxy benzaldehyde

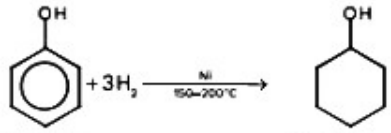
(ix) Mercuration : Phenol when heated with mercuric acetate undergoes mercuration to form o-and p-isomers.



o-HydroxyPhenyl mercuricacetate

p-Hydroxyphenyl mercuricacetate

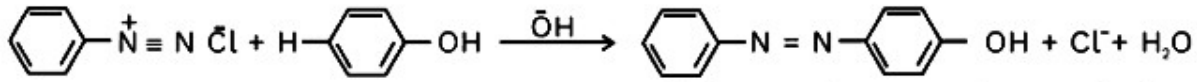
(x) Hydrogenation: Phenol, when hydrogenated in presence of a nickel catalyst at about 150-200°C, forms cyclohexanol.



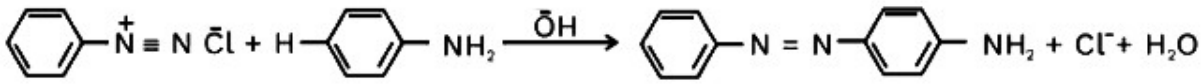
Phenol (C_6H_5OH)

Cyclohexanol ($C_6H_{11}OH$) (used as a good solvent)

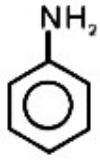
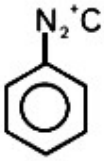
(xi) Coupling Reaction:

(I) 

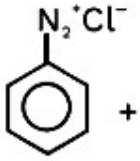
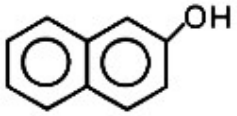
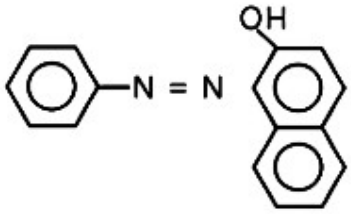
p-Hydroxyazobenzene (orange dye)

(II) 

p - Aminoazobenzene (yellow dye)

(III)  + $NaNO_2$ + 2HCl $\xrightarrow{0°-5°C}$  + NaCl + $2H_2O$

Benzene diazonium chloride

 +  $\xrightarrow{0°-5°C}$  + HCl

β-Naphthol (2-Naphthal)

Naphthol aniline dye (Orange-red dye)